

Lecture 3: Decision Theory under Uncertainty

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Uncertainty in Decision Making

- Virtually every decision is made under uncertainty
- Examples:
 - Saving for retirement with uncertain income, returns, health
 - Firms choosing products and prices under uncertain demand
 - Governments choosing policies under uncertain growth, inflation, unemployment

Objectives of the Course

- Develop a model of choice behavior under uncertainty
- Start with von Neumann-Morgenstern (vN-M) expected utility model
- Study:
 - Measures of risk aversion
 - Ranking uncertain prospects
 - Comparative statics under uncertainty

Limitations of Expected Utility

- Assumes objective risk with known probabilities
- Example: coin flips
- But many real-world cases involve subjective uncertainty
- Savage (1954): derives subjective probabilities from choice behavior
- Later: behavioral criticisms of expected utility

Prizes and Consequences

- Prize space X : set of possible outcomes
- Example: $X = \{\text{Stanford wins, Cal wins, Tie}\}$
- Economic applications: usually monetary payoffs, but not always

- Lottery $p = (p_1, \dots, p_n)$ is a probability distribution over X
- $\Delta(X) = \{(p_1, \dots, p_n) : p_i \geq 0, \sum_i p_i = 1\}$
- Example: $p = (1/2, 1/2, 0)$, $p' = (5/9, 3/9, 1/9)$
- Represent lotteries in probability simplex

Convexity of Lotteries

- Given $p, p' \in \Delta(X)$ and $\alpha \in [0, 1]$
- $\alpha p + (1 - \alpha)p' \in \Delta(X)$
- Interpretation: convex combination $\Rightarrow$ compound lottery
- First-stage lottery decides which p or p' applies, then outcome realized

Compound Lotteries

- Example: outcome depends first on quarterback choice, then game result
- Two-stage process:
 - Stage 1: choose p with prob α , p' with prob $1 - \alpha$
 - Stage 2: play chosen lottery
- Mathematically: $\alpha p + (1 - \alpha)p'$

Probability Simplex

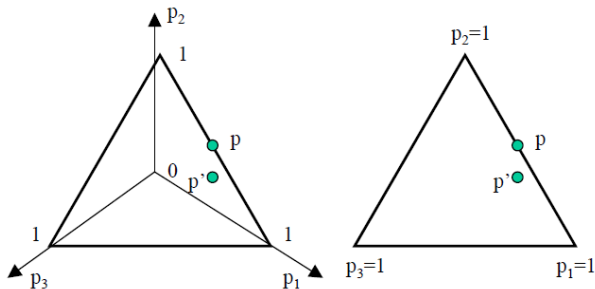


Figure 1: The Probability Simplex

Compound Lottery

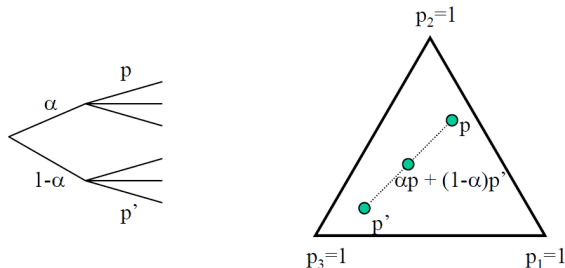


Figure 2: Figure 2: A Compound Lottery

Preferences over Lotteries

- Decision-maker has preferences $\succeq$ over lotteries $P = \Delta(X)$
- Assume: preferences are complete and transitive
- Add two more axioms: continuity and independence

Definition

A preference relation $\succeq$ on P is **continuous** if for any $p, p', p'' \in P$ with

$$p \succeq p' \succeq p'',$$

there exists $\alpha \in [0, 1]$ such that

$$\alpha p + (1 - \alpha)p'' \sim p'.$$

- Preferences vary smoothly
- If $p \succ p'$, then a lottery close to p is still preferred to p'

Definition

A preference relation $\succeq$ satisfies **independence** if for all $p, p', q \in P$ and $\alpha \in [0, 1]$,

$$p \succeq p' \iff \alpha p + (1 - \alpha)q \succeq \alpha p' + (1 - \alpha)q.$$

- If $p \succ p'$, this holds even when both are mixed with the same q
- Also called the *substitution axiom*

Example: Consumer Theory vs. Independence

- Suppose I prefer $\{2 \text{ cokes}, 0 \text{ twinkies}\}$ to $\{0 \text{ cokes}, 2 \text{ twinkies}\}$
- This does *not* imply that I prefer

$\{2 \text{ cokes}, 1 \text{ twinkie}\}$ to $\{1 \text{ coke}, 2 \text{ twinkies}\}$

- Even though these are “averages” of the first two bundles with $\{2 \text{ cokes}, 2 \text{ twinkies}\}$
- Consumer preferences over goods do not obey independence
- Independence is specific to uncertainty, where mixing preserves preferences

- Continuity: no sudden jumps in preferences
- Independence: captures consistent trade-offs under uncertainty
- Independence has no direct analogue in standard consumer theory

Expected Utility Form

Definition

A utility $U : P \rightarrow \mathbb{R}$ has an **expected utility** form if $\exists (u_1, \dots, u_n)$ s.t.

$$U(p) = \sum_{i=1}^n p_i u_i, \quad p \in \Delta(X).$$

- u_i : utility of certain outcome x_i
- Linear in probabilities

Linearity Property

- For any p, p' and $\alpha \in [0, 1]$:

$$U(\alpha p + (1 - \alpha)p') = \alpha U(p) + (1 - \alpha)U(p').$$

- Linearity $\iff$ expected utility form
- Key: indifference sets are affine

Ordinal vs. Cardinal

- Consumer theory: any increasing transform preserves prefs
- Expected utility: only *affine* transforms preserve prefs
- Form: $V(p) = a + bU(p)$, $b > 0$

Proposition 1 (Affine Uniqueness)

Statement

If U is an expected utility representation of $\succeq$, then V is also an expected utility representation of $\succeq$ iff

$$V(p) = a + bU(p), \quad b > 0.$$

- Uniqueness up to location and scale
- Affine transforms keep rankings of lotteries

Proof Sketch: “If” Direction

- Suppose $V(p) = a + bU(p)$, $b > 0$
- If $U(p) \geq U(p')$ then $V(p) \geq V(p')$
- V has expected utility form: set $v_i = a + bu_i$
- Then $V(p) = \sum_i p_i v_i$

Proof Sketch: “Only If” Direction (1/2)

- Assume U and V are EU representations of the same $\succsim$
- Let $p^* = \text{best lottery}$, $p_* = \text{worst lottery}$ (exist since X finite)
- Assume $p^* \succ p_*$
- For any $p \in P$, continuity $\Rightarrow \exists \lambda_p \in [0, 1]$ such that

$$U(p) = \lambda_p U(p^*) + (1 - \lambda_p) U(p_*)$$

and

$$p \sim \lambda_p p^* + (1 - \lambda_p) p_*.$$

Proof Sketch: “Only If” Direction (2/2)

- By the same argument for V :

$$V(p) = \lambda_p V(p^*) + (1 - \lambda_p) V(p_*).$$

- Thus

$$V(p) = a + bU(p), \quad a = V(p_*) - \frac{U(p_*)}{U(p^*) - U(p_*)} (V(p^*) - V(p_*)),$$

$$b = \frac{V(p^*) - V(p_*)}{U(p^*) - U(p_*)}.$$

- Hence V is an affine transform of U

Theorem 1 (vN–M, 1947)

A complete and transitive preference $\succeq$ on lotteries P satisfies continuity and independence $\iff$ it admits an expected utility representation $U : P \rightarrow \mathbb{R}$.

- Core justification for expected utility theory
- Independence $\iff$ parallel indifference curves

- 3 outcomes $\Rightarrow$ lotteries in 2D simplex
- Expected utility $\Rightarrow$ indifference curves are parallel straight lines
- If curves not parallel $\Rightarrow$ violates independence

Figure: Parallel Indifference Curves

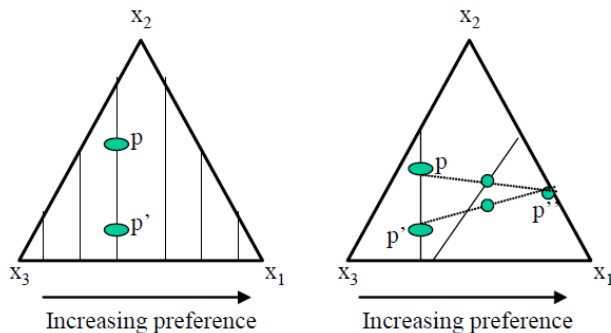


Figure 3: Independence implies parallel linear indifference curves

Proof Sketch: Step 1

- Assume $\succeq$ is complete, transitive, continuous, independent
- Let p^* = best lottery, p_* = worst lottery (exist since X finite)
- By continuity + independence: if $1 > \beta > \alpha > 0$, then

$$p^* \succ \beta p^* + (1 - \beta)p_* \succ \alpha p^* + (1 - \alpha)p_* \succ p_*.$$

Proof Sketch: Step 2

- For any lottery p , $\exists! \lambda_p \in [0, 1]$ such that

$$p \sim \lambda_p p^* + (1 - \lambda_p) p_*.$$

- Define $U(p) = \lambda_p$
- Normalization: $U(p^*) = 1, U(p_*) = 0$

Proof Sketch: Step 3

- U represents preferences:

$$p \succeq q \iff U(p) \geq U(q).$$

- Check linearity:

$$p \sim U(p)p^* + (1 - U(p))p_*, \quad q \sim U(q)p^* + (1 - U(q))p_*.$$

Proof Sketch: Step 4 (Linearity)

- Independence $\Rightarrow$

$$\alpha p + (1 - \alpha)q \sim (\alpha U(p) + (1 - \alpha)U(q))p^* + (1 - \alpha U(p) - (1 - \alpha)U(q))p_*.$$

- So

$$U(\alpha p + (1 - \alpha)q) = \alpha U(p) + (1 - \alpha)U(q).$$

- Hence U has expected utility form

- Proof relies on finite X (best/worst exist)
- For infinite X : need extra axiom (Sure Thing Principle)
- Establishes expected utility as unique representation of axioms

Setup: Money Lotteries

- Prize space $X = \text{real numbers}$ (money outcomes)
- Lottery represented by cdf $F(x)$
- Expected utility representation:

$$U(F) = \int u(x) dF(x)$$

where $u = \text{Bernoulli utility function}$

- Assume u is increasing and continuous

Definition: Risk Aversion

Definition

A decision-maker is **risk-averse** if for any non-degenerate lottery F with mean $E[F]$,

$$\delta_{E[F]} \succeq F,$$

where $\delta_{E[F]}$ is the degenerate lottery giving $E[F]$ for sure.

- Strict risk aversion: strict preference for the mean over the risky lottery
- Captures dislike of uncertainty

Jensen's Inequality

- Risk aversion $\iff$

$$\int u(x) dF(x) \leq u\left(\int x dF(x)\right).$$

- This is Jensen's inequality
- Characteristic of concave u

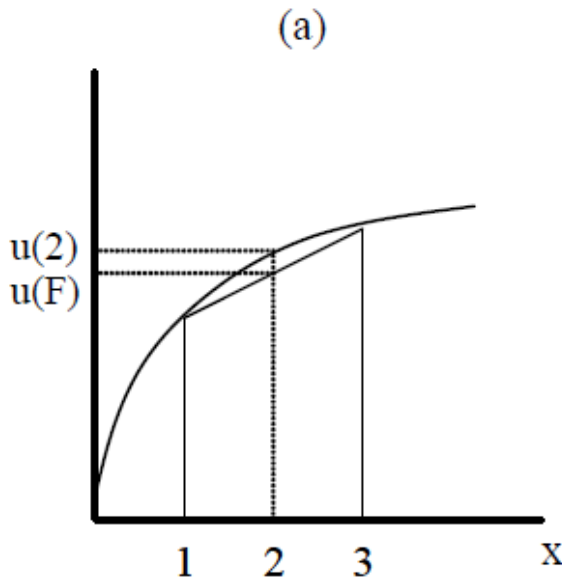
Proposition 2

Proposition

A decision-maker is risk-averse $\iff$ the Bernoulli utility function u is concave.

- Concavity of u = diminishing marginal utility of wealth
- Linear u : risk-neutral
- Convex u : risk-loving

Figure: Illustration of Risk Aversion



Definition

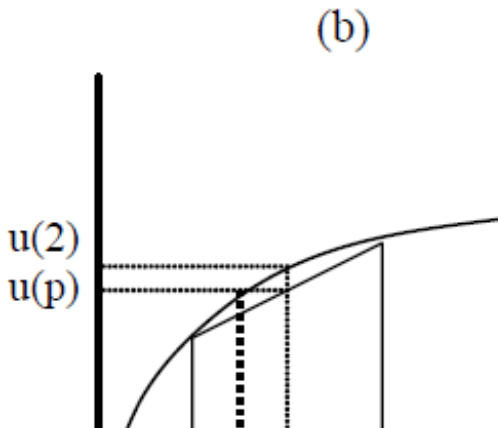
The **certain equivalent** $c(F, u)$ of lottery F satisfies

$$u(c(F, u)) = \int u(x) dF(x).$$

- Value of risky lottery expressed in certain dollars
- If u is concave, $c(F, u) \leq E[F]$
- Measures willingness to accept risk

Graphical Representation

- Figure 4(b): certain equivalent $c(F, u)$ is left of mean $E[F]$ under concavity
- Captures degree of risk aversion



Definition

For u twice differentiable, the **coefficient of absolute risk aversion** is

$$A(x) = -\frac{u''(x)}{u'(x)}.$$

- Captures local curvature of u
- Higher $A(x) \Rightarrow$ greater risk aversion at wealth x

Proposition 3 (Pratt, 1964)

Equivalent Definitions of “More Risk-Averse”

Utility u is more risk-averse than $v \iff$ any of the following hold:

- 1 If u prefers lottery F to sure x , then so does v
- 2 For all F : $c(F, u) \leq c(F, v)$
- 3 $\exists$ increasing concave g with $u = g \circ v$
- 4 $\forall x$: $A(x, u) \geq A(x, v)$

- (1) Preference-based: u turns down more lotteries than v
- (2) Certain equivalent smaller for u than v
- (3) u is “more concave” than v
- (4) Arrow–Pratt: local curvature higher for u

- How do risk preferences change with wealth?
- Example: Bill Gates vs. average person
- If I accept a risky gamble, Gates should also accept (he is wealthier)
- Leads to notion of decreasing absolute risk aversion (DARA)

Definition

The Bernoulli utility u has:

- Decreasing absolute risk aversion (DARA) if $A(x) = -\frac{u''(x)}{u'(x)}$ decreases in x
- Constant absolute risk aversion (CARA) if $A(x)$ is constant
- Increasing absolute risk aversion (IARA) if $A(x)$ increases in x
- Intuition: richer agents are more willing to take fixed-dollar risks under DARA

Implications of DARA

- If $F \succ x$ at wealth w
- Then $F \oplus z \succ x + z$ at higher wealth $w + z$
- $\Rightarrow$ willingness to accept risk grows with wealth

Proportionate Gambles

- Gamble pays random multiple t of wealth x
- Certain equivalent for rate of return t :

$$u(\hat{t}x) = \int u(tx) dF(t).$$

- $\hat{t}$ = certain rate of return equivalent

Definition

Coefficient of relative risk aversion:

$$R(x) = -x \frac{u''(x)}{u'(x)}.$$

- Measures aversion to proportional gambles
- DRRA: $R(x)$ decreasing in x
- Intuition: richer agents willing to risk same fraction of wealth

Proposition 4

Statement

Agent exhibits decreasing relative risk aversion (DRRA) $\iff$ the certain equivalent $\hat{t}$ of a proportional gamble increases with wealth x .

- DRRA $\Rightarrow$ richer agents accept same gamble on larger scale
- Relation: $R(x) = xA(x)$
- DRRA $\Rightarrow$ DARA (but not vice versa)

Interpretation

- DARA: absolute willingness to bear risk grows with wealth
- DRRA: proportional willingness to bear risk grows with wealth
- Intuitive: 20at1% of wealth vs. 20at0.01% of wealth

Motivation

- How to compare different lotteries F and G ?
- Need criteria that work across all decision-makers with monotone/concave u
- Key tools: stochastic dominance (first and second order)

First Order Stochastic Dominance (FOSD)

Proposition 5

Distribution G first-order stochastically dominates F if

$$G(x) \leq F(x), \forall x,$$

or equivalently, for all nondecreasing u :

$$\int u(x) dG(x) \geq \int u(x) dF(x).$$

- G always yields weakly better outcomes than F
- Preferred by all agents who like “more to less”

FOSD: Intuition

- Statistical view: G more likely to pay above any threshold x
- Preference view: all monotone u prefer G
- Equivalent to improving outcomes via a compound lottery (upward shifts)

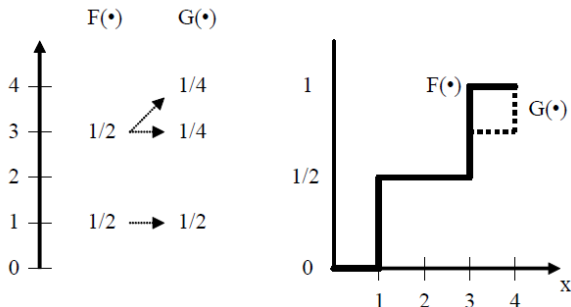


Figure 6: Figure 5: First Order Stochastic Dominance Shift

Second Order Stochastic Dominance (SOSD)

Proposition 6

Let F, G have same mean. G second-order stochastically dominates F if

$$\int_{-\infty}^x G(y) dy \leq \int_{-\infty}^x F(y) dy, \quad \forall x,$$

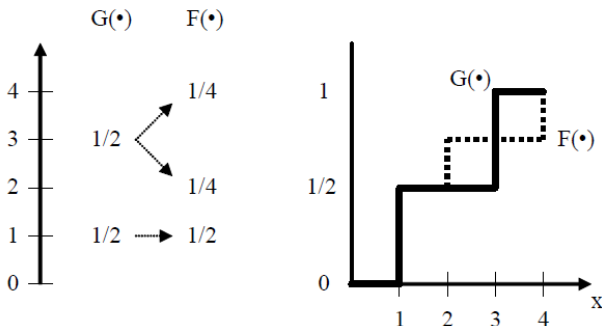
or equivalently, for all concave u :

$$\int u(x) dG(x) \geq \int u(x) dF(x).$$

- G is “less risky” than F with same expected payoff
- Preferred by all risk-averse agents

Mean-Preserving Spreads

- Rothschild–Stiglitz (1970): riskier lottery = mean-preserving spread
- Construct F from G by adding mean-zero noise
- F more “spread out” but same mean
- $\text{SOSD} \iff F$ can be obtained from G via mean-preserving spreads



- FOSD: one lottery always better (all monotone agents agree)
- SOSD: one lottery less risky with equal mean (all risk-averse agents agree)
- Provides robust criteria for ranking risky prospects

The Demand for Insurance: Setup

- Agent with wealth w
- Faces probability p of loss L
- Buys insurance coverage a at price q per unit payout
- Optimization:

$$\max_a pu(w - qa - L + a) + (1 - p)u(w - qa)$$

First Order Condition

- Differentiate w.r.t. a :

$$pu'(w - qa - L + a)(1 - q) - (1 - p)u'(w - qa)q = 0$$

- Condition: marginal benefit in loss state = marginal cost in no-loss state

Actuarially Fair Insurance

- Fair if $q = p$
- Then FOC simplifies:

$$u'(w - qa - L + a) = u'(w - qa)$$

- Solution: $a = L$ (full insurance)
- Risk-averse agent always fully insures under fair prices

Insurance Above Fair Price

- If $q > p$, full insurance not optimal
- Solution $a^* < L$
- Interpretation: transferring wealth to loss state is costly
- Comparative statics:
 - With DARA: insurance demand decreases with wealth
 - With CARA/IARA: different patterns

Portfolio Choice: Setup

- Wealth w
- Two assets:
 - Safe: return r
 - Risky: return $z \sim F$
- Investment a in risky, $w - a$ in safe
- Final wealth: $az + (w - a)r$
- Optimization:

$$\max_a \int u(az + (w - a)r) dF(z)$$

First Order Condition

- Differentiate w.r.t. a :

$$\int u'(a(z - r) + wr)(z - r) dF(z) = 0$$

- Risk-neutral ($u(x) = x$): invest all in asset with highest mean return
- Risk-averse ($u'' < 0$): interior solution if $E[z] > r$

Interpretation

- Risk-averse agents still invest in risky asset if $E[z] > r$
- At $a = 0$, marginal benefit of risky asset positive if $E[z] > r$
- So optimal $a^* > 0$ even for most risk-averse

Connection to Insurance

- Insurance problem $\sim$ portfolio problem
- Buying full insurance $\equiv$ investing only in safe asset
- If insurance priced above fair value, not optimal to fully insure
- By same logic, some risk is always taken if it yields positive expected return

Subjective Probability: Motivation

- vN–M: assumes objective probabilities (coin flips)
- Real world: many events lack objective probabilities
- Example: Microsoft stock up? Global warming impact?

Savage's Theorem (1954)

- Acts: functions $f : S \rightarrow X$ mapping states to outcomes
- If preferences satisfy axioms (incl. sure-thing principle), then:

$$f \succeq g \iff \sum_{s \in S} u(f(s))p(s) \geq \sum_{s \in S} u(g(s))p(s)$$

- Interpretation: subjective probabilities $p(s)$ derived from choices
- Foundation for Bayesian decision theory

Allais Paradox (1953)

- **Problem 1:** Choose
 - A: 55,000 (0.33), 48,000 (0.66), 0 (0.01)
 - B: 48,000 for sure
- **Problem 2:** Choose
 - C: 55,000 (0.33), 0 (0.67)
 - D: 48,000 (0.34), 0 (0.66)
- Typical: B in Problem 1, C in Problem 2
- Violates independence axiom (preferences inconsistent)
- Explained by **certainty effect**: overweighting sure outcomes

Rabin's Calibration Theorem (2000)

- EU model: risk aversion $\iff$ concave u
- Example: reject 50/50 lose 1000/gain 1050
- Implies: must also reject 50/50 lose 1000/gain 1 billion!
- Concavity cannot explain moderate risk aversion
- Suggests alternative: **loss aversion** (losses loom larger than gains)

Ellsberg Paradox (1961)

- Urn with 300 balls: 100 red, 200 either white or blue (unknown)
- **Bet 1:** Win \$100 if guess right (guess red or white)
- **Bet 2:** Win \$100 if guess wrong (guess red or white)
- Common choice: red in both bets
- Inconsistent with subjective probability model
- People prefer known risks over unknown risks (**ambiguity aversion**)

Framing Effects (Asian Disease Problem)

- Disease expected to kill 600 people
- Programs:
 - A: 200 saved for sure
 - B: 1/3 prob 600 saved, 2/3 prob none saved
 - C: 400 die for sure
 - D: 1/3 prob none die, 2/3 prob 600 die
- Results: majority choose A over B, but D over C
- $A=C$ and $B=D \Rightarrow$ preferences flip under framing
- Highlights role of reference points and framing in choices

Summary of Criticisms

- Allais paradox: violations of independence axiom
- Rabin: concavity implausible explanation of risk aversion
- Ellsberg: ambiguity aversion contradicts Savage
- Framing: reference dependence and heuristics matter
- Led to new theories: prospect theory, ambiguity models